

STUDENT PERFORMANCE ANALYTICS SYSTEM

Project Documentation

1. Objective

The objective of this project is to develop a Python-based **Student Performance Analytics System** that can process student academic records and generate useful performance information.

The system reads student records from a CSV file and uses Python, Pandas and NumPy to calculate total marks, average marks, grades and pass/fail status.

The project also analyzes the dataset to identify the top-performing student, calculate subject-wise averages, determine the best-performing subject, and count the number of passed and failed students.

The project demonstrates the practical use of Python fundamentals, functions/processing logic, loops, conditions, NumPy and Pandas for data analysis.

2. Technologies Used

Python

Python is used as the main programming language for implementing the Student Performance Analytics System. Python concepts used include variables, lists, conditional statements, for loops, DataFrame operations, indexing and data processing.

Pandas

Pandas is used for reading, storing and analyzing the student dataset. The project uses Pandas to read the CSV file, create and manipulate a DataFrame, calculate total and average marks, find maximum values, calculate subject-wise averages, and filter and analyze student records.

NumPy

NumPy is imported and used as part of the numerical data-analysis implementation. It supports numerical operations used during processing.

3. Dataset Description

The project uses a CSV dataset named **Students_Records.csv**. The dataset contains student academic information.

Column	Description
Student_ID	Unique identification number of the student
Name	Name of the student
Maths	Marks obtained in Mathematics
Physics	Marks obtained in Physics
Chemistry	Marks obtained in Chemistry
Attendance	Student attendance percentage

The three subjects analyzed by the program are Maths, Physics and Chemistry. The dataset is loaded into a Pandas DataFrame using `pd.read_csv('Students_Records.csv')`. The final documentation should state the exact number of records present in the submitted CSV file.

4. Implementation

Step 1 – Upload the Dataset

The notebook provides an option to upload the required dataset using the Google Colab file-upload functionality.

Step 2 – Read the Data

Pandas is used to load the CSV file into a DataFrame. The first few records are displayed using `df.head()` to inspect the data.

Step 3 – Define Subjects

The subjects Maths, Physics and Chemistry are defined and stored in a Python list for performance calculations.

Step 4 – Calculate Total Marks

The total marks of each student are calculated using `df[subjects].sum(axis=1)`.

Step 5 – Calculate Average Marks

The average marks are calculated using `df[subjects].mean(axis=1)`.

Step 6 – Assign Grades

A for-loop processes every student's average and assigns a grade according to the implemented grading rules: A+ for 90 or above, A for 80–89.99, B for 70–79.99, C for 60–69.99, D for 50–59.99, and F below 50.

Step 7 – Determine Pass/Fail Status

A student is classified as Pass when Maths, Physics and Chemistry are each at least 40 and attendance is at least 75%. Otherwise, the student is classified as Fail.

Step 8 – Display Student Performance

The project displays Student ID, Name, Total, Average, Grade and Pass/Fail status.

Step 9 – Identify the Top Student

The student with the highest average is identified using `df['Average'].idxmax()`.

Step 10 – Calculate Subject-Wise Performance

The program calculates the average marks for each subject and identifies the subject with the highest average.

Step 11 – Calculate Pass/Fail Statistics

The program counts the number of passed and failed students and displays the overall results.

5. Key Features

- CSV student dataset handling.
- Pandas DataFrame creation and processing.

- Calculation of total marks.
- Calculation of average marks.
- Automatic grade assignment.
- Pass/fail classification.
- Attendance-based pass condition.
- Identification of the highest-performing student.
- Calculation of subject-wise averages.
- Identification of the best-performing subject.
- Calculation of the number of passed students.
- Calculation of the number of failed students.
- Use of loops and conditional statements.
- Use of Pandas for data analysis.
- Use of NumPy for numerical-analysis support.

6. Output / Screenshots

Screenshot 1 – Dataset

Add a screenshot showing the student dataset after it has been loaded into the Pandas DataFrame.

[Insert Screenshot Here]

Figure 1: Student dataset loaded using Pandas.

Screenshot 2 – Student Performance Summary

Add a screenshot showing Student ID, Name, Total, Average, Grade and Pass/Fail.

[Insert Screenshot Here]

Figure 2: Calculated student performance summary.

Screenshot 3 – Overall Analysis

Add a screenshot showing the top student, highest average, subject-wise averages, best-performing subject, passed students and failed students.

[Insert Screenshot Here]

Figure 3: Overall student performance analysis.

7. Final Outcome

The Student Performance Analytics System successfully processes student academic records and produces meaningful performance analysis. The system calculates individual students' total and average marks, assigns grades, determines pass/fail status using marks and attendance, identifies the highest-performing student, and analyzes subject-wise performance.

The project demonstrates how Python, Pandas and NumPy can be used to transform raw student records into useful information. The final output makes it easier to understand individual and overall class performance without manually calculating every result.

8. Challenges & Learning

Challenges

- Reading and organizing the student dataset correctly.
- Performing calculations for multiple subjects.
- Applying different grading conditions correctly.
- Implementing multiple conditions for pass/fail classification.
- Finding the highest-performing student from the calculated averages.
- Performing subject-wise analysis.

How the Challenges Were Solved

These problems were handled using Pandas DataFrame operations, Python for loops, conditional statements and built-in data-analysis functions such as `sum()`, `mean()`, `max()` and `idxmax()`.

Learning Outcomes

- Reading CSV data using Pandas.
- Working with DataFrames.
- Performing numerical calculations.
- Using Python lists and loops.
- Using conditional statements.
- Creating calculated DataFrame columns.
- Performing subject-wise data analysis.
- Finding maximum values from a dataset.
- Understanding practical applications of NumPy and Pandas.

Conclusion

The Student Performance Analytics System is a Python-based data-analysis project designed to analyze student academic performance. The project demonstrates the use of Python programming fundamentals along with Pandas and NumPy to process student records, calculate performance metrics, assign grades, determine pass/fail status, and identify important performance trends.